Customer Churn in Telecom: Predictive Modeling for Enhanced Retention Strategies

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Agenda

1. **Problem Statement**
   Understand customer churn. Assess its critical business impact.
2. **Project Objectives**
   Define project goals. Outline key motivations.
3. **Context & Value**
   Review prior solutions. Highlight our unique value creation.
4. **Data Analysis**
   Extract key insights. Analyze the churn dataset.
5. **Methodology & Design**
   Prepare data. Develop robust predictive models.
6. **Implementation**
   Design web app frontend. Build scalable backend.
7. **Evaluation**
   Assess model performance. Ensure system reliability.
8. **Next Steps & Conclusion**
   Plan future enhancements. Summarize project impact.

Introduction

- **Overview:**
  - Addressing customer churn in Egypt's telecom sector using data-driven approaches.
  - Predictive model development to identify customers at risk of churning.
  - Deployment of results via a web application.
- **Why it Matters:**
  - Telecom providers face increasing churn rates, impacting revenue and market reputation.
  - Predictive analytics offer opportunities for proactive retention strategies.

Problem Statement

Definition of Churn: Customers discontinuing services with a telecom provider.

- **Core Challenges:**
  - High acquisition costs vs. retention costs.
  - Competition-driven customer migration.
  - Insufficient tools for early churn prediction in Egypt's market.

Impact: Revenue loss, brand damage, and operational inefficiencies.

Project Motivation

- **Key Drivers:**
  - Retaining customers is up to 5x cheaper than acquiring new ones.
  - Churn prediction helps identify dissatisfied customers before they leave.
  - AI and machine learning have proven to enhance predictive capabilities.

Unique Value: Tailoring solutions to Egypt's telecom market using localized data.

Project Objectives

1. **Develop an ML Model:**
   - Predict customers at high risk of churning.
2. **Actionable Insights:**
   - Highlight key drivers of churn for targeted strategies.
3. **User-Friendly Interface:**
   - Enable non-technical teams to leverage predictive insights.
4. **Visual Analytics:**
   - Provide churn trends and key metrics in an intuitive format.
5. **Scalability:**
   - Ensure the solution adapts to growing datasets and market needs.

Related Work

- **Machine Learning Techniques:**
  - Random Forest (86.94% accuracy) and Gradient Boosting.
  - Strength: Simplicity and interpretability.
- **Deep Learning Advances:**
  - ChurnNet (97.52% accuracy).
  - Strength: Capturing complex patterns but computationally expensive.
- **Hybrid Models:**
  - Ensemble methods like XGBoost and LightGBM (96.65% accuracy).
  - Combines the best of multiple algorithms.

Business Model Overview

Value Proposition: Predictive analytics to reduce churn, improve customer satisfaction, and drive profitability.

Customer Segments: Telecom companies in Egypt, decision-makers, and analysts.

Channels: User-friendly web app with real-time analytics and downloadable reports.

Customer Relationships: Self-service platform, personalized support, and regular updates.

Revenue Streams: Subscriptions, setup fees, and consultancy services.

Key Resources: Machine learning models, Kaggle dataset, and development team.

Key Activities: Data processing, model refinement, and user support.

Key Partnerships: Data providers, academic advisors, and telecom stakeholders.

Cost Structure: Development, maintenance, and operational expenses.

Methodology

- **Data Collection:**
  - Source: IBM Kaggle Telco dataset.
  - Content: Demographics, usage patterns, service details.
- **Preprocessing:**
  - Handle missing values, encode categorical variables, normalize data.
- **Model Training:**
  - Algorithms tested: Random Forest, Logistic Regression, ANN.
- **Deployment:**
  - Interactive web application for real-time predictions.

Overview of IBM Kaggle Telcom dataset

Context:

- The dataset pertains to a fictional telco company providing home phone and Internet services to 7,043 customers in California.
- The data includes 33 variables capturing various customer attributes and behaviors.

Key Variables:

Demographics: Gender, Senior Citizen status, Partner, Dependents.

Service Subscriptions: Phone Service, Multiple Lines, Internet Service, Online Security, Online Backup, Device Protection, Tech Support, Streaming TV, Streaming Movies.

Contract Details: Contract type, Paperless Billing, Payment Method.

Financials: Monthly Charges, Total Charges.

Churn Information: Churn Label, Churn Value, Churn Score, Churn Reason, CLTV (Customer Lifetime Value).

Data Analysis Highlights

1. **Churn Overview:**
   - Most customers stay, but a notable percentage churn, highlighting retention challenges.
2. **Key Drivers of Churn:**

Demographics: Senior citizens and customers without partners are more likely to churn.

Services: Lack of phone service, Fiber Optic internet, and missing add-ons like online security increase churn risk.

Contracts: Month-to-month contracts have higher churn compared to longer-term contracts.

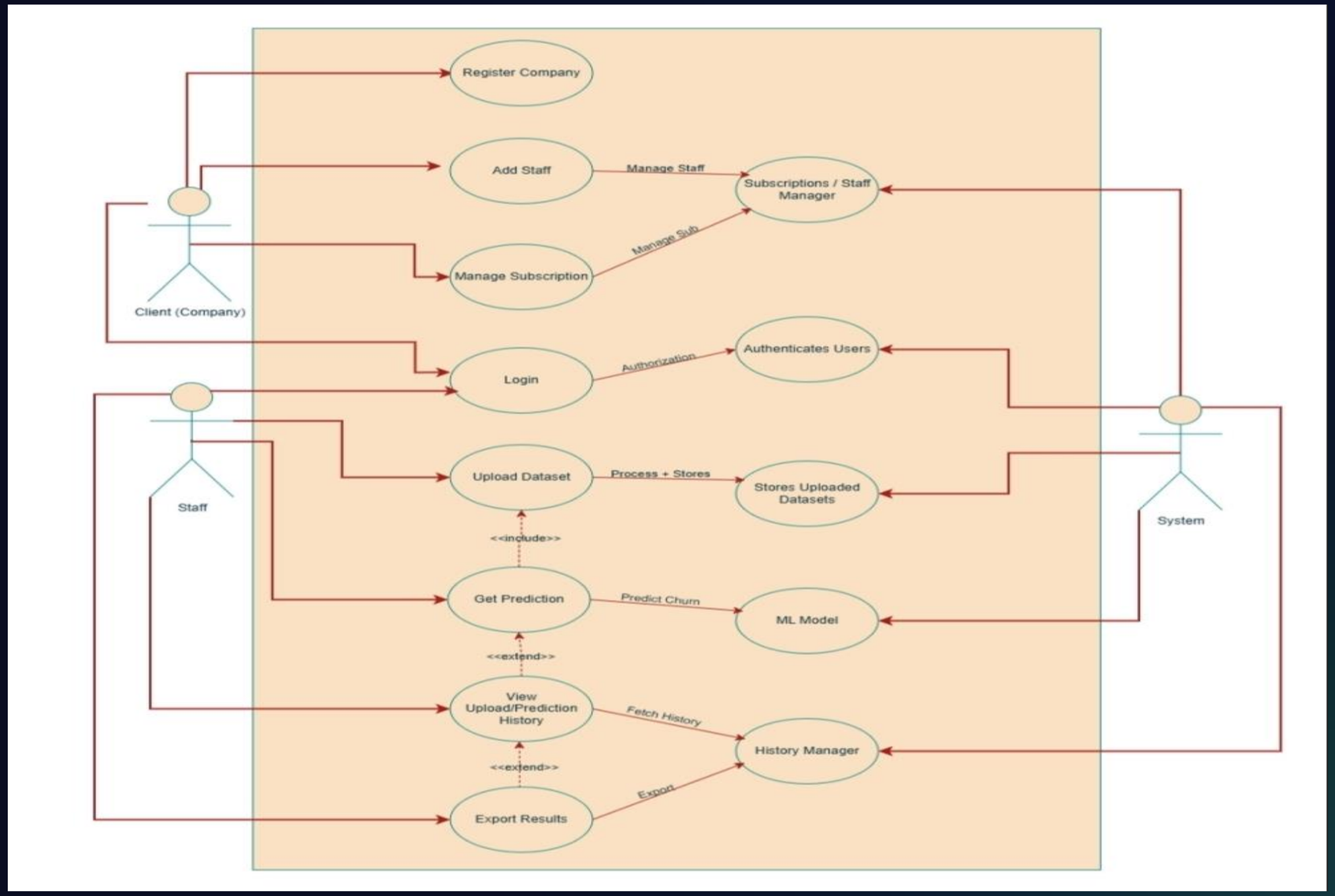
Financials: Higher monthly and total charges correlate with increased churn.

3. **Positive Retention Factors:**
   - Customers with dependents, longer tenure, and those subscribing to additional services (e.g., online backup) are less likely to churn.

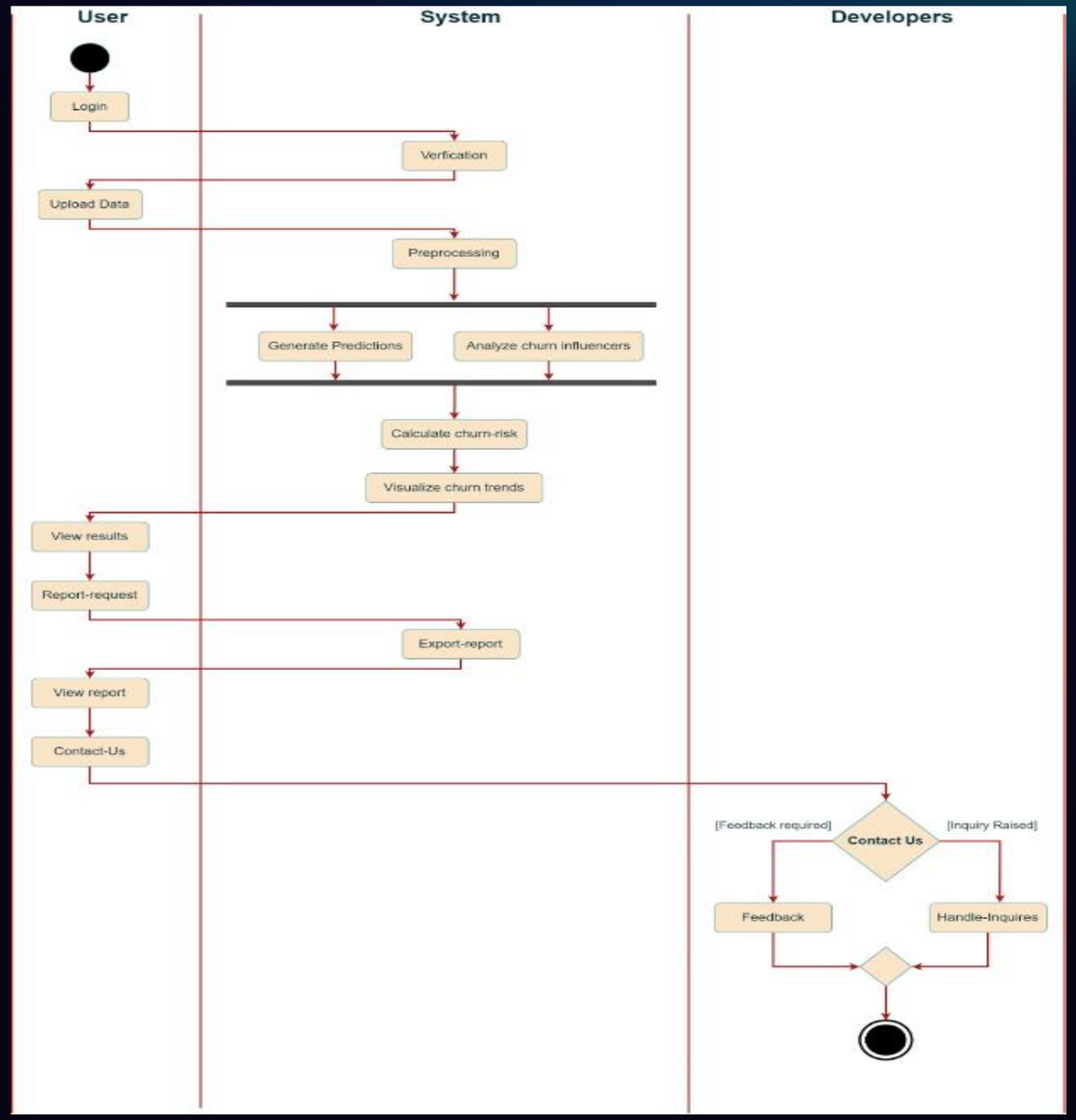
Analysis and Design

- Diagrams:
  - Use Case Diagram
  - Activity Diagram
  - ER Diagram
  - Data Flow Diagrams
    - Context diagram
    - Level 0 diagram
    - Level 1 diagram
  - Class diagram
  - Sequence diagram

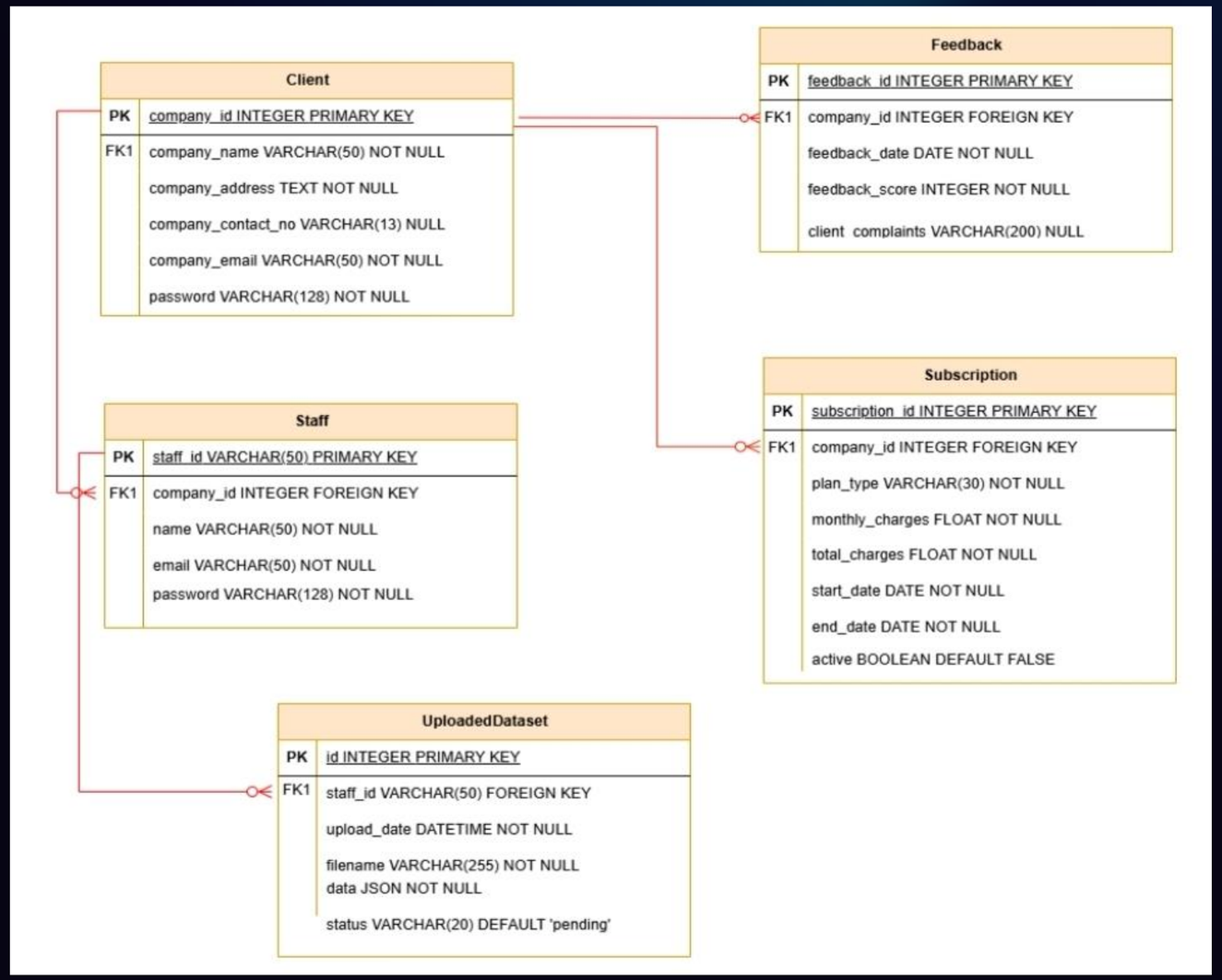
Use Case Diagram



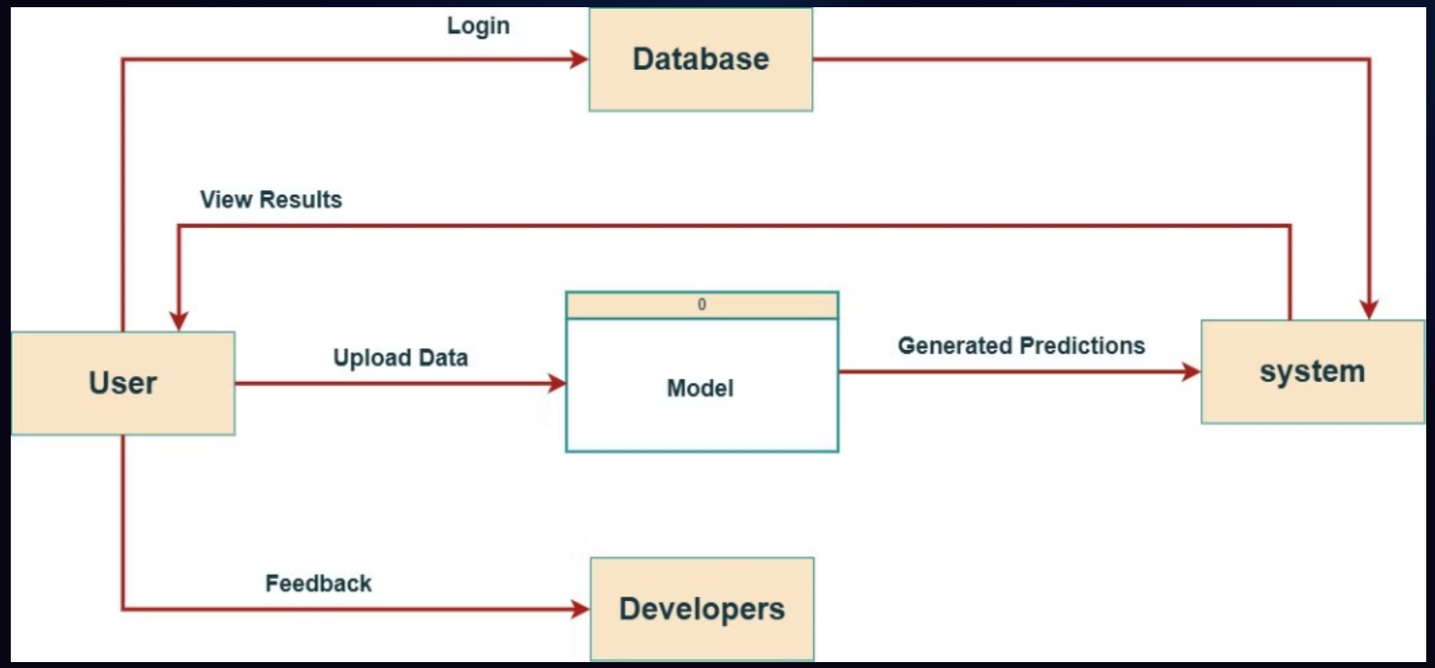
Activity Diagram



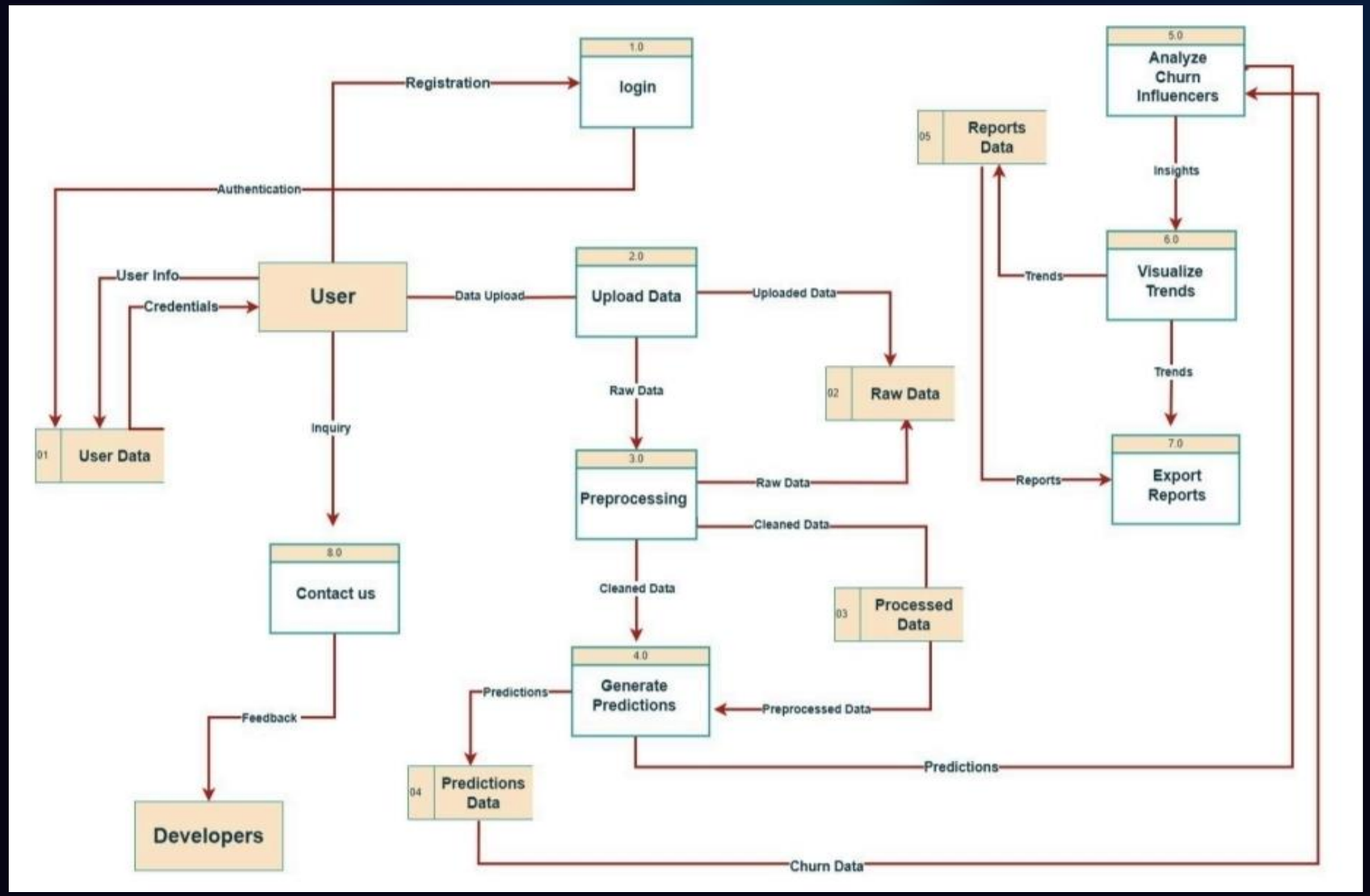
Entity Relationship Diagram (ERD)



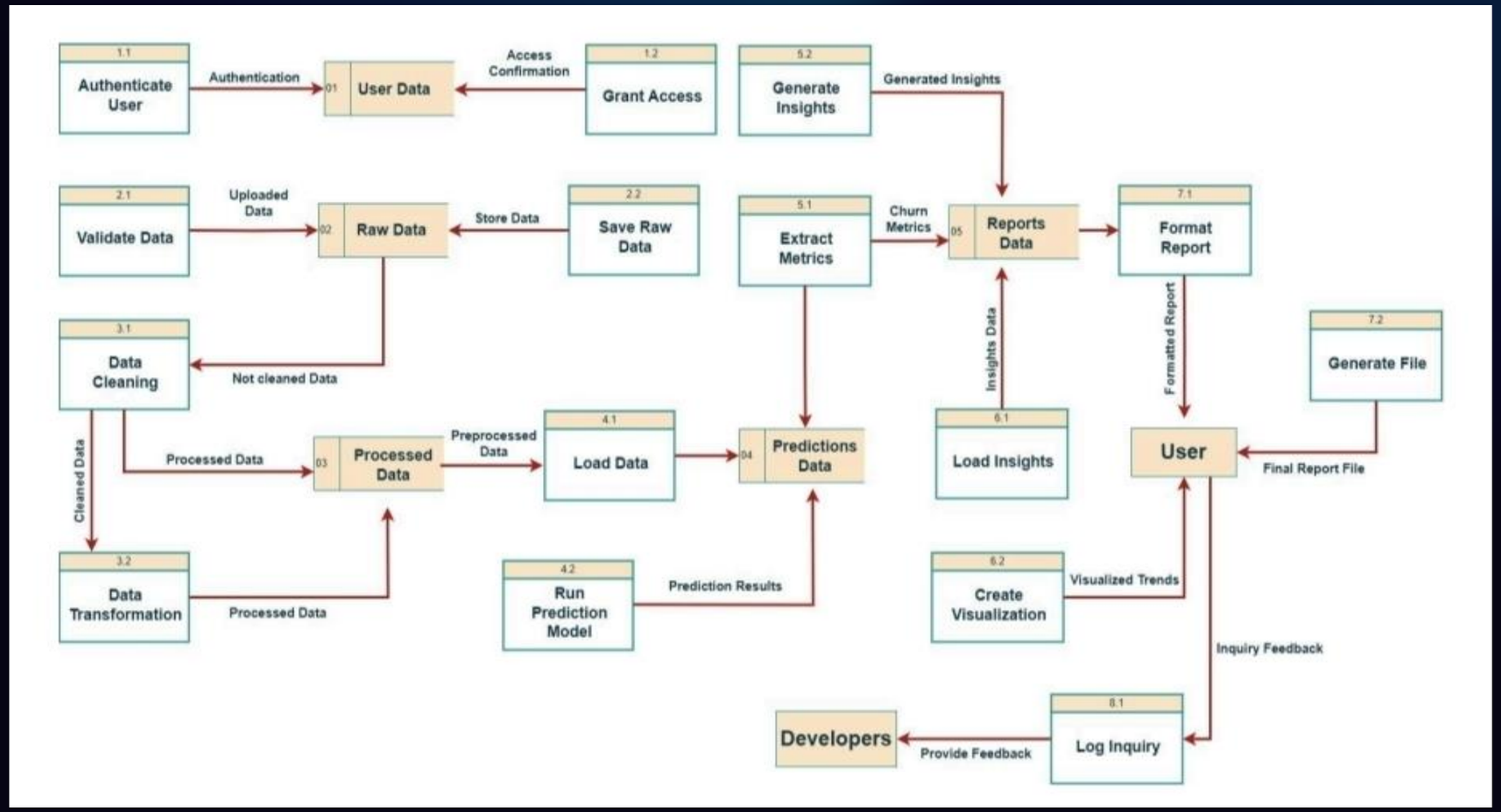
Data Flow Context diagram



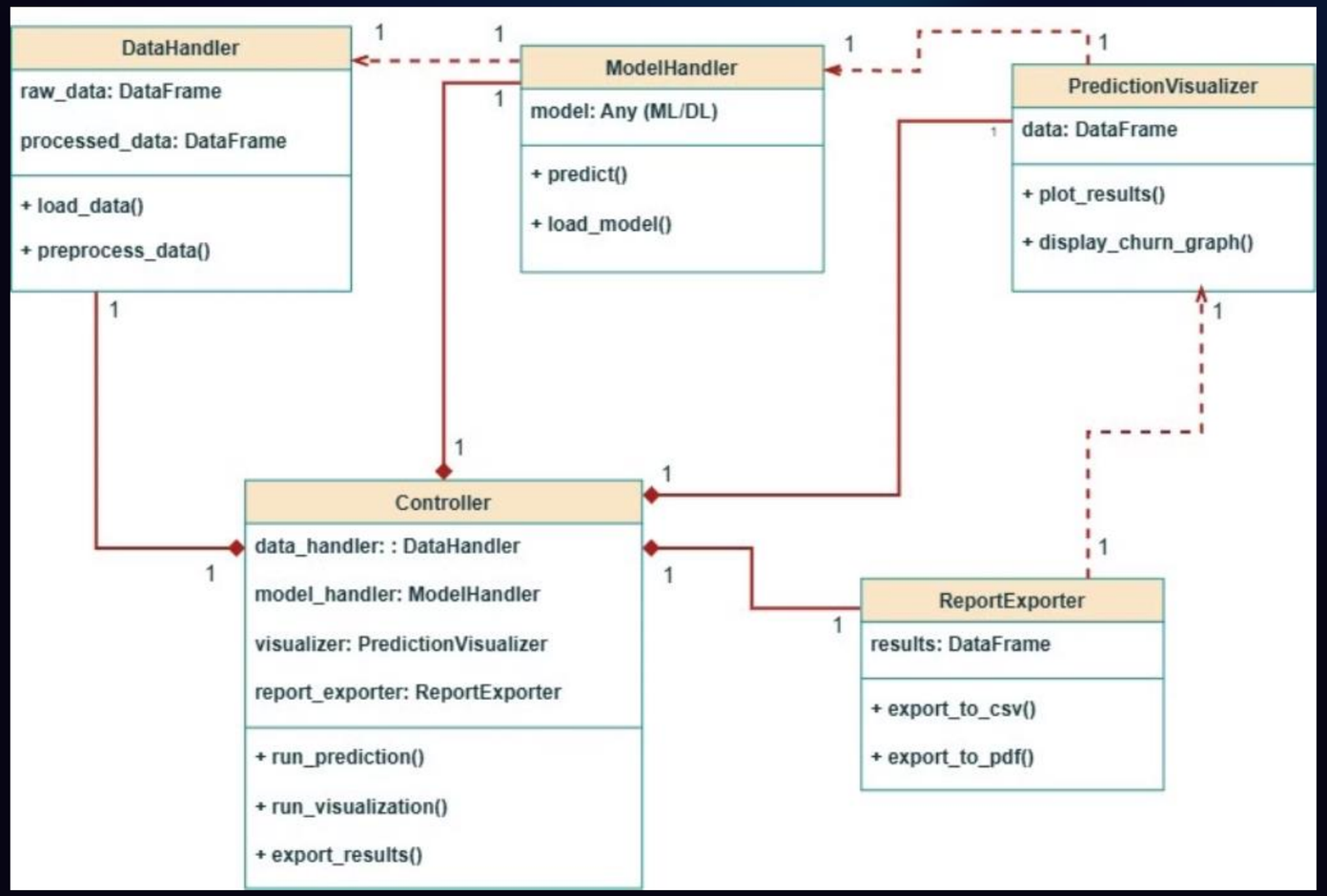
Data Flow Level 0 diagram



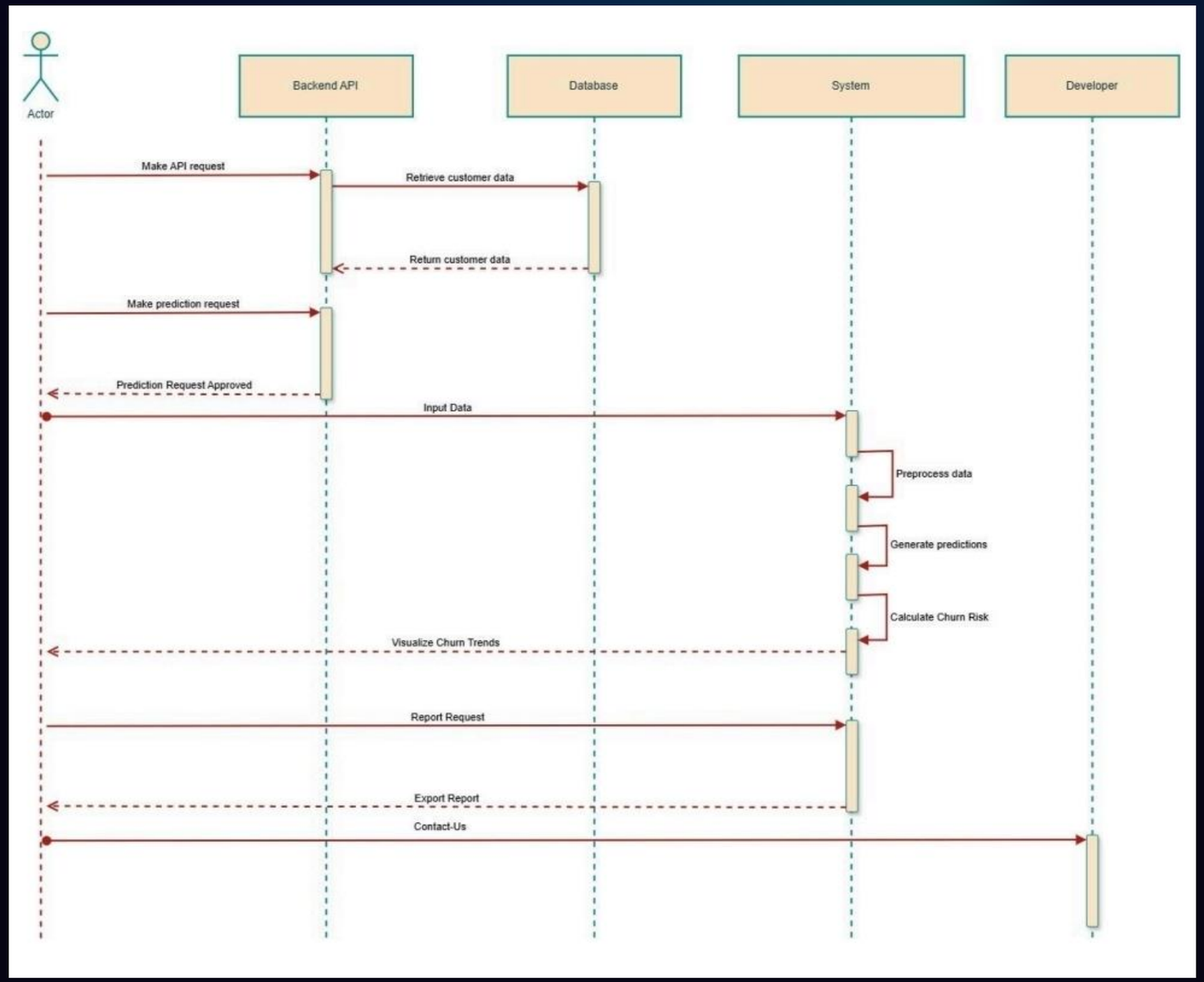
Data Flow Level 1 diagram



Class diagram



Sequence diagram



Functional Requirements

- **Data Handling:**
  - Accepts bulk uploads (CSV).
- **Prediction Engine:**
  - Identifies customers with high churn probability.
- **Insights:**
  - Feature importance rankings (e.g., contract type, billing issues).
- **Visualizations:**
  - Churn trends, customer segmentation, and risk distribution charts.

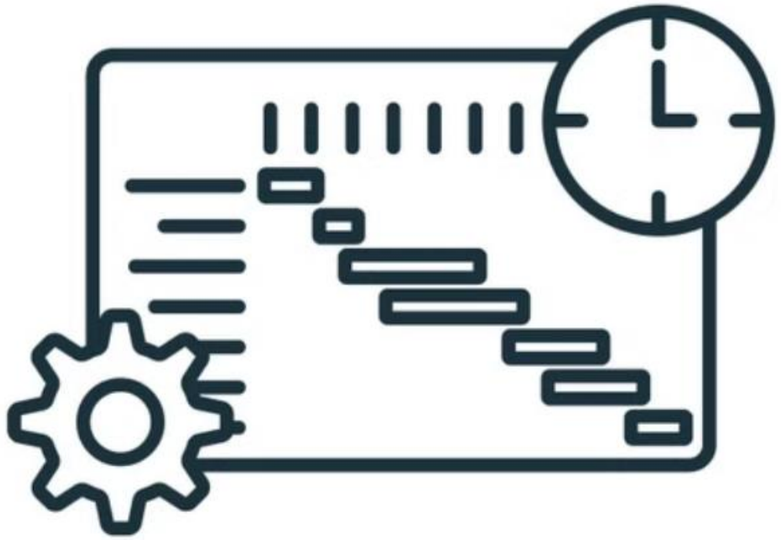
Non-Functional Requirements

Accuracy: Minimum of 85% for reliable predictions.

Performance: Response time under 5 seconds for batch processing.

Scalability: Handles datasets with 100,000+ records.

Security: Ensures no unauthorized access to sensitive customer data.



TIMELINE

Implementation Timeline

Phase 1: October–December 2024

- Literature review, problem analysis, and system design.

Phase 2: January–February 2025

- Model development and preparing for developing web application.

Phase 3: March–April 2025

- Web application development and integration.

Phase 4: May 2025

- Testing and finalizing documentation and presenting results.

Methodology : Data Preparation & Processing

Dataset

- Source: IBM Telco Churn
- Size: 7,000+ customer records
- Features: demographics, services, contracts, billing

Preprocessing

- Handled missing values
- Label encoded categorical features
- Normalized numerical val

Methodology: Model Development & System Deployment

Machine Learning Models

Advanced machine learning models evaluated include:

- XGBoost
- LightGBM
- Random Forest
- Decision Tree

Final Model: Stacking Ensemble

Model Performance:

- Cross-Validation Score: 91.2%
- ROC-AUC: 84.5%

System Deployment

- **Backend:** Django-powered backend with REST APIs for robust data management and prediction serving.
- **Frontend:** Intuitive user interface developed using React and Tailwind CSS.
- **Security:** Robust authentication secured by JSON Web Tokens (JWT).
- **API Endpoints:** Facilitating real-time individual churn predictions and efficient batch data exports.

System Architecture & Backend Implementation

Our system seamlessly integrates a robust backend with an intuitive frontend. This full-stack design drives powerful churn prediction capabilities.

The Django backend efficiently manages core data models and ensures secure access using JWT authentication.

Integrated machine learning pipelines, leveraging a serialized ensemble model, enable real-time predictions.

Frontend, Workflow & Extensibility

Frontend Functionality

React-based interface with Tailwind styling

Key features:

- Secure login
- Data upload (CSV)
- Trigger churn prediction
- Export results (CSV/Excel)

System Workflow

Staff logs in

Uploads cust

Model Evaluation & Performance

Cross-Validation (CV)

Checks model consistency across diverse data subsets for robust performance.

ROC-AUC Score

Measures the model's ability to distinguish between churners and non-churners effectively.

Confusion Matrix

Analyzes true and false predictions, providing detailed insights into model accuracy.

XGBoost	90.02%	82.40%
LightGBM	88.02%	80.40%
Random Forest	85.64%	77.23%
Decision Tree	84.23%	76.53%
Ensemble (Final)	**91.23%**	**84.53%**

The stacking ensemble model was chosen for its superior accuracy and ROC-AUC score. It reliably identifies at-risk customers, enhancing retention strategies.

System Testing & Validation

Unit Testing

Tests individual backend components in isolation for correctness.

Integration Testing

Verifies seamless connections across API, frontend, and ML components.

End-to-End Testing

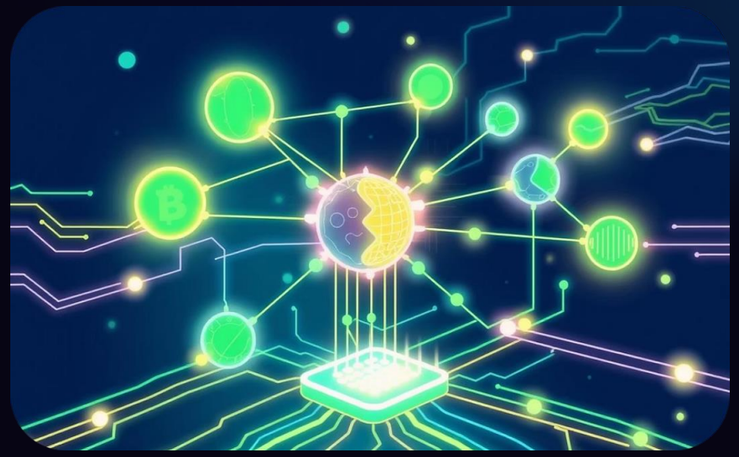
Validates comprehensive user scenarios and complete workflows.

Key test scenarios and their outcomes:

Login	Staff login	Redirect to dashboard	Pass
Data Upload	Upload CSV	Data preview shown	Pass
Prediction	Trigger model	Results displayed	Pass
Export	Export results	CSV file downloaded	Pass
Invalid Login	Wrong credentials	Error message shown	Pass

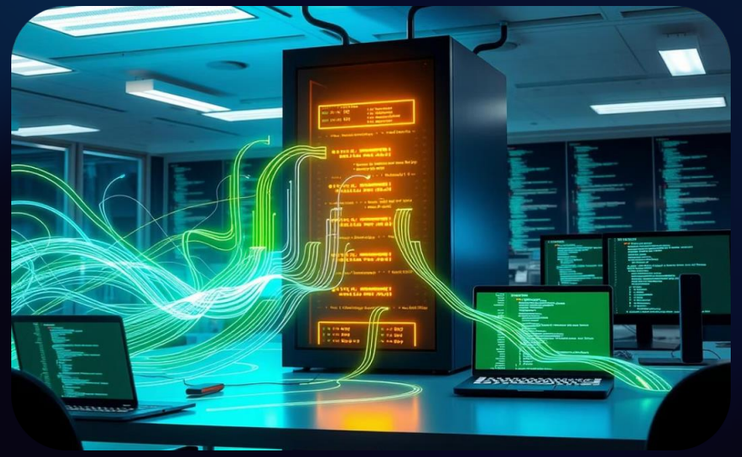
- All critical system paths have been thoroughly tested and successfully validated.
- The system is confirmed reliable, secure, and ready for deployment.

Future Work: Strategic Enhancements



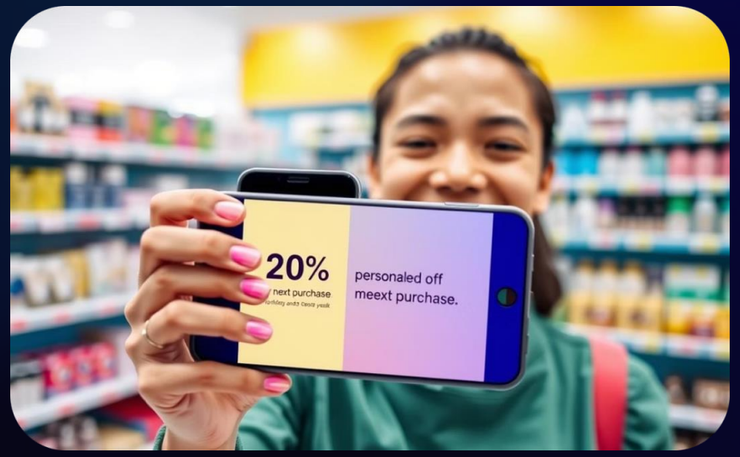
Automated Model Retraining

Periodically retrain the model with newly uploaded data to maintain accuracy and relevance.



Real-Time Data Integration

Connect to live data sources like CRM systems for immediate insights and proactive churn prevention.



Retention Recommendations

Suggest personalized actions (e.g., discounts, upgrades) based on prediction results to maximize customer retention.

Conclusion: Project Success & Impact

Accurate Predictions

TeleChurn predicts customer churn with high accuracy.

Robust Full-Stack System

Developed with Django, React, and seamless ML integration.

Strong Performance

Achieved 91.2% CV and 84.5% ROC-AUC scores.

Ready for Deployment

Secure, user-friendly, and prepared for real-world application.

Significant Business Value

Helps telecom companies retain customers and reduce revenue loss.

Questions and Answers

This concludes our presentation on the Telecom Churn Prediction System. We appreciate your time and welcome any questions you may have.